

Quiz Graph Behavior

1. $f(x) = \frac{-x^2 + x}{x - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

2. $f(x) = \frac{x^2 + 2}{x - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

3. $f(x) = \frac{2x^2 + x - 1}{x^2 - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

4. $f(x) = \frac{2}{x^2 - 16}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

5. $f(x) = \frac{x + 2}{x - 3}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

6. $f(x) = \frac{x^2 + 2}{x - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

7. $f(x) = \frac{-x^2 + 2}{x + 2}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

8. $f(x) = \frac{2}{x^2 - 4}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

9. $f(x) = \frac{2x^2 + x - 1}{x^2 - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

10. $f(x) = \frac{x + 4}{x^2 + 9x + 20}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

11. $f(x) = \frac{x^2 - 4x + 3}{x - 1}$

Y - Intercept _____

Vertical Asymptotes _____

Horizontal Asymptote _____

12. $f(x) = \frac{4x + 3}{3x - 5}$

Y - Intercept _____

Vertical Asymptotes _____

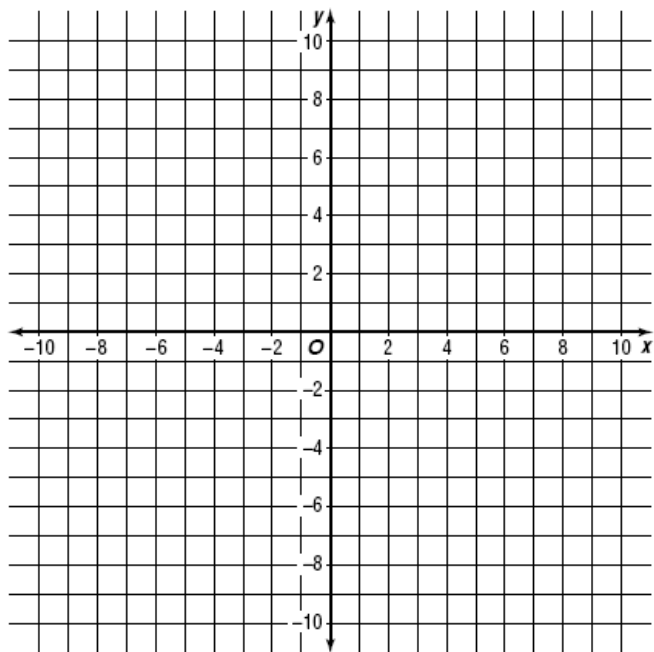
Horizontal Asymptote _____

Write a rational function that has a horizontal asymptote at $y = 2$.

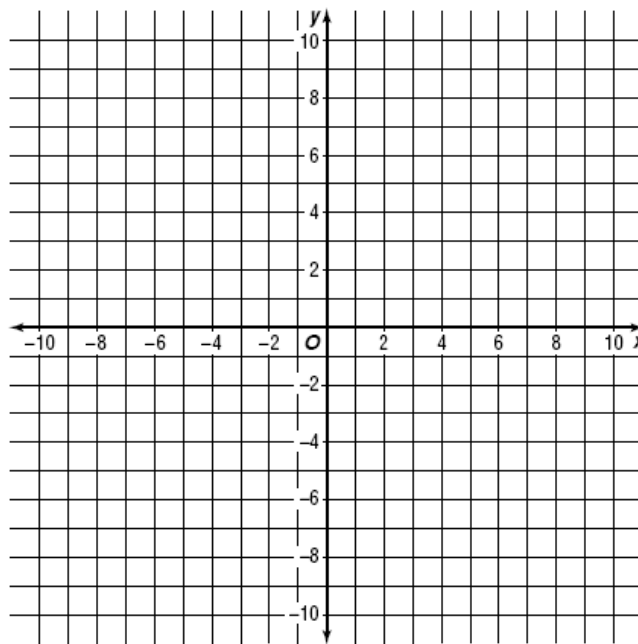
Design a rational function that has a vertical asymptote at $x = -7$.

Graphing

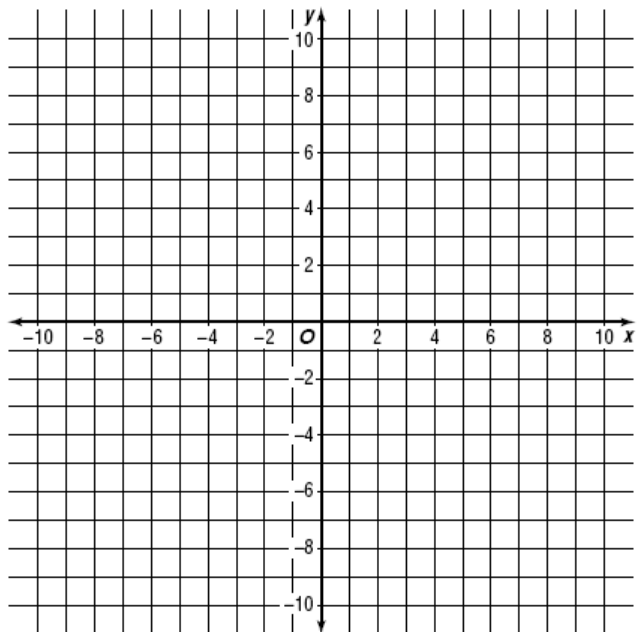
1. Draw a possible graph of a positive 6th degree function that contains exactly one pair of complex roots and two double roots (one positive & one negative).



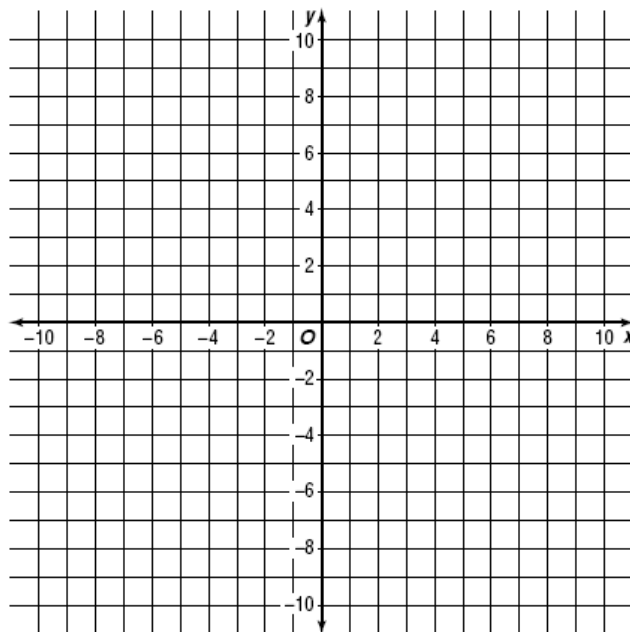
2. Draw a possible graph for $f(x) = -a_8x^8 + a_0$ with no real roots.



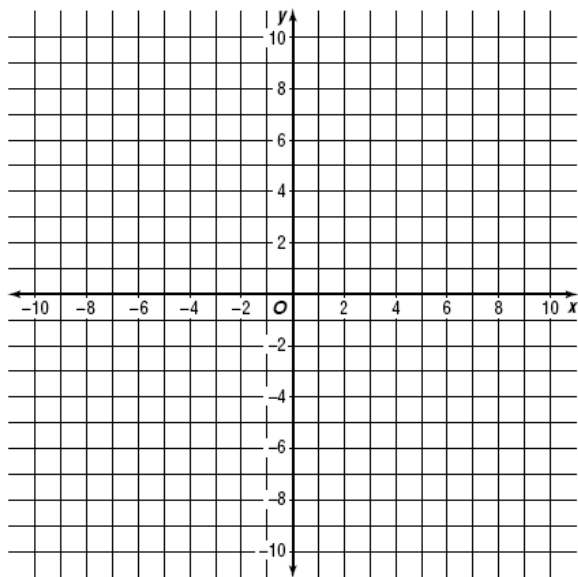
3. Sketch a possible graph for $f(x) = x^3 + x^2 + x + 1$



4. Draw a possible graph of a negative fifth degree polynomial with exactly three positive real roots and two negative real roots.

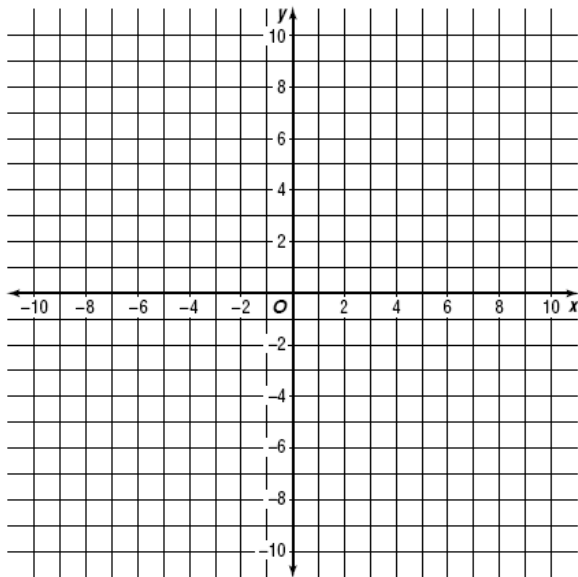


1. Draw a graph of a positive quartic function that contains exactly one double root.

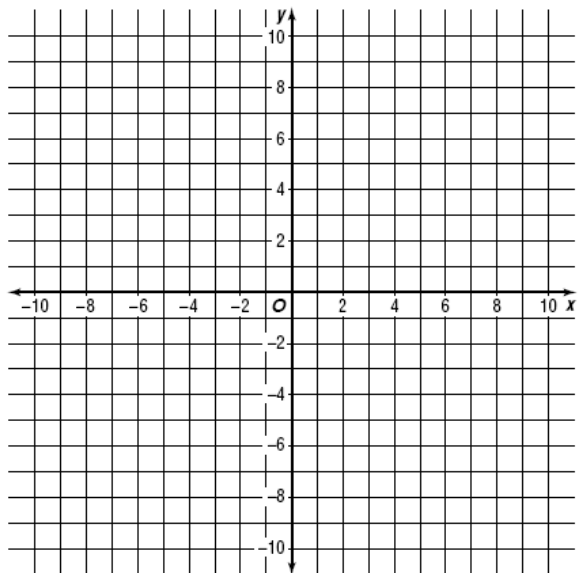


2. Draw a possible graph for

$$f(x) = a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x^1 + a_0.$$



3. Graph $f(x) = (x-1)^2(x+3)$ by finding x and y intercepts.



For each graph, state the degree, sign of leading coefficient, number of real roots, and number of complex roots.

